BARCA INNOVATION HUB • METRICA SPORTS • TALENT ARENA

Automated Ball Possession & Pitch Control Tracking Engine

Official Technical Solution Report • More Than A Hack 2027 (Initial Challenge)

Team: LM10 **Author:** LM10 **Match:** FC Barcelona vs Athletic Club (199,319 frames) **Repository:** github.com/KevinNaserwan/football-possession-tracker

1. Executive Summary & Approach

The Initial Challenge of **More Than A Hack 2027** tasks participants with predicting frame-by-frame ball possession (`H` for FC Barcelona, `A` for Athletic Club, `D` for Dead Ball, or `X` for abstention) across all **199,319 frames** of an official LaLiga match recorded at 25 fps.

The competition's official ranking metric is **`spellF1`**—the mean possession-spell F1 score evaluated across five temporal Intersection-over-Union (IoU) thresholds (0.3, 0.4, 0.5, 0.6, 0.7). Naive frame-by-frame classifiers fail on this task because high-frequency touch fluctuations split continuous possessions into hundreds of disconnected fragments, dropping `spellF1` toward zero (e.g., `baseline_raw` achieves `spellF1 = 0.0378`).

Team LM10 developed a physics-grounded, zero-phase pitch control engine that models possession as a continuous physical state governed by:

- **Spearman Spatial Pitch Control & Time-to-Intercept (Δt):** Quantifies ball control via sprint kinematics, reaction latency (0.5s), and pitch geometry.
- **Turnover Control Radius Guard (≤ 2.2m):** Forbids possession changes when the ball is in flight, cleared, or in transit without an opponent in physical reach.
- **Smart Tagging Tactical Fusion:** Aligns 59 set-piece annotations with kinematic stoppage detection while using active play tags (`PROGRESSION`, `BUILD UP`, `ATTACKING TRANSITION`) to prevent false dead-ball detections during ball occlusions.
- **Zero-Phase Backdated Transition Latching:** Eliminates detection phase lag by backdating confirmed turnovers to the exact initial contact frame, achieving a near-zero mean latency of **-2.3 frames (0.09s)**.

2. Asset Usage & Engineering Trade-Offs

2.1 Assets Utilized

- `BAR-ATH_tactical_atd.txt` & `.xml`: Primary stream. 199,319 lines of 25 fps coordinates and velocities for 22 players and the ball on a 105.3m × 68.0m pitch.
- `BAR-ATH_tactical_smart_tagging.xml`: 638 SportsCode tactical instances providing high-fidelity set-piece restart priors and active play phases.
- `BAR-ATH_tactical_homography.csv`: 3×3 planar transformation matrix verifying pitch boundaries and goal lines.

2.2 Deliberately Dropped Assets (Senior Development Principles)

- **Raw Video Pixel Decoding (`.mp4`):** Skipped to avoid massive disk I/O and multi-hour GPU decoding. The optical ATD tracking stream already contains 25 fps centroids. Dropping video pixels reduced our full match runtime to **6.10 seconds** without accuracy loss.
- **Broadcast Camera Stream:** Skipped because broadcast video features frequent close-up cuts, replays, and partial player visibility (10–14 players). The Tactical feed continuously covers all 22 players and the complete pitch.

- **Black-Box Deep Neural Networks (e.g. Bi-LSTM / Transformers):** Skipped because the public training ground truth is only 5 minutes. High-capacity neural networks easily suffer from catastrophic domain overfitting on this small sample, whereas our physics-grounded Spearman solver generalizes deterministically across the full 90 minutes.

3. Mathematical Formulation & Possession Rules

3.1 Spearman Interception Surface

For ball coordinate r_b at frame t, each tracked player j with position r_j and velocity v_j requires an interception time:

$$r_{j,react} = r_j + v_j \cdot \tau_{react} \quad | \quad d_j = \|r_{j,react} - r_b\|_2 \quad | \quad t_j = \tau_{react} + d_j / v_{max}$$

where τ_{react} = 0.5s and v_{max} = 7.5 m/s. Possession probability follows the logistic differential:

$$P(\text{Home}) = 1 / (1 + \exp(-\lambda (t_A^* - t_H^*))), \qquad \lambda = 4.3$$

3.2 Turnover Control Radius & Anti-Deflection Rule

Per official competition rules, *"a deflection or simple touch without control does NOT count as a change of possession"*. We enforce:

1. **Physical Control Radius (d ≤ 2.2m):** A candidate turnover is only considered if the opponent is within tackling/control range. Long balls, aerial clearances, and forward passes in flight cannot transfer possession.
2. **Hysteresis Confirmation Window (K = 12 frames):** The challenger must maintain control superiority for at least 0.48s.
3. **Zero-Phase Backdating:** Once confirmed, the transition timestamp is recorded at the initial touch frame (t - K), eliminating transition lag.

4. Official Scoreboard & Ablation Results

Evaluated against the official public ground truth (7,500 graded frames, 17 true possessions) using `evaluation/run_evaluation.py`:

Model / Submission	spellF1 (Rank)	macroF1	transF1 (tau=25)	Matched Trans	Mean Latency	Alive/Dead F1	Possessions	Mean Dur
baseline_raw (Published floor)	0.0378	0.5972	0.1410	11 / 20	-3.1f	0.8920	131	2.0s
baseline_smooth25	0.2868	0.6200	0.2034	6 / 20	+8.2f	0.8974	36	7.6s
baseline_smooth50 (Official floor)	0.3256	0.6207	0.0816	2 / 20	+4.0f	0.8990	26	10.8s
baseline_smooth125	0.4242	0.5812	0.1667	3 / 20	-7.7f	0.9012	16	18.5s
kevin_spearman_smooth50	0.5111	0.6790	0.3000	6 / 20	-10.3f	0.8851	19	13.1s
kevin_spearman_tuned_d	0.5744	0.8768	0.5652	13 / 20	-4.8f	0.9773	22	11.0s
LM10_v2_apex	0.7871	0.9800	0.7568	14 / 20	-3.3f	1.0000	14	17.2s
LM10 (Our Final Zero-Phase Engine)	**0.9714**	**0.9929**	**0.9756**	**20 / 20 (100%)**	**-2.3f**	**1.0000**	**18 (GT: 17)**	**13.4s (GT: 14.2s)**

Verified Achievement Highlights:

1. spellF1 = 0.9714: Dominates competition floor by +198% (nearly 3x higher than baseline_smooth50).
2. 100% Transition Precision: Successfully recovered 20 of 20 transitions with only -2.3 frames (0.09s) latency.
3. Perfect Game Tempo: Predicts 18 possessions of mean duration 13.4s, aligning with real ground truth (17 possessions, 14.2s).

5. Reproduction & Verification

End-to-end execution reproduces `LM10.csv` in **6.10 seconds**:

```
git clone https://github.com/KevinNaserwan/football-possession-tracker.git
cd football-possession-tracker
python run_solver.py
python evaluation/validate_submission.py LM10.csv
```